COE202 Mathematical Foundations of AI

Spring 2026

Course Project

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1 Overview

As part of this course, you will complete a project related to your research interests. The goal is to explore a mathematical or AI-related topic in depth, formulate the problem clearly, and develop a meaningful technical solution and presentation.

The final project is worth 20% of your final grade. Projects should be completed in **teams of 1–3 students**, so please **find teammates if needed before submitting your initial proposal**.

The topic of your project is intentionally flexible, as long as it is connected to mathematics, machine learning, optimization, probability, or related topics covered in this course. You may consider one of the following types of projects:

- Conduct an original practical research project by applying machine learning methods to a specific problem or dataset. Interdisciplinary problems and applications are especially welcome.
- Conduct an original theoretical research project motivated by open problems in machine learning or AI. You have already seen some of these problems in the "Aha!" topics discussed in class.
- Critically revisit a published paper by reproducing its methods, experiments, or theoretical analysis.
- Write a literature review on a specific topic in machine learning and provide a critical comparison of existing methods (ideally supported by experiments on a standard benchmark dataset).

2 Deliverables

This project consists of two stages of deliverables:

- **Initial Proposal (due Monday, May 25)**
- **Final Report, Presentation Video and Slides (due Thursday, June 11)**

Each deliverable is described in more detail in the following sections.

3 Initial Proposal

Please submit a **1–2 page** proposal describing your project. Your proposal should include:

- **Motivation:** What problem or research question are you interested in?

- **Preliminary Work (optional):** Briefly describe any related work, experiments, or ideas you have already explored.
- **Proposed Work:** What do you plan to do in the project? What methods, models, algorithms, datasets, or theoretical ideas will you investigate?
- **Expected Outcome:** What do you hope to learn, demonstrate, compare, or achieve?

After completing the proposal, **all team members** should submit it to the BB system.

4 Final Report

Your final report should include the following sections:

- **Problem Setup**
  - problem statement;
  - mathematical formulation.
- **Technical Approach**
  - methods, algorithms, or theoretical ideas;
  - experimental setup (if applicable).
- **Main Results**
  - theoretical and/or experimental results;
  - figures, tables, or analysis.
- **Conclusion**
  - summary of findings;
  - limitations and future directions.

There is no minimum length requirement, but the report must **not exceed 8 pages**. This is a strict requirement. After completing the report, **all team members** should submit it to the BB system.

5 Presentation Video and Slides

As part of the final project, you are required to submit a presentation video together with your slides. Please follow the guidelines below:

- The video should clearly explain the main aspects of your project, including the motivation, problem setup, methods, and main results.
- The presentation should be self-contained. It is recommended to present the main ideas and results using concise mathematical and visual explanations.
- There is no required video format, but the video must be **no longer than 10 minutes**, including a **one-sentence introduction for each collaborator**.

After completing the video, **only one submission per team is required**. Please send the video and slides together **by email** (yaow@sustech.edu.cn and 12231275@mail.sustech.edu.cn), and cc your collaborators if applicable. There is **no need for each team member to submit separately**.

6 Grading Criteria

The final report and presentation video will be evaluated based on the following criteria:

- **Consistency:** Your writing, mathematical formulation, experiments, and presentation should be logically consistent throughout the project.
- **Relevance:** Your project should be related to mathematics, machine learning, AI, optimization, probability, or related topics covered in this course.
- **Usefulness:** Your project should investigate an interesting, meaningful, or insightful question.
- **Clarity:** Your report and presentation should be clear, well-organized, and easy to follow.

7 Project Ideas

7.1 Benefits of Depth in Neural Networks

Understanding the expressive power and trainability of neural networks is one of the central problems in deep learning. For example:

- How does network architecture affect expressivity and optimization?
- Can the performance of a neural network architecture be predicted theoretically?
- How do depth, width, skip connections, normalization layers, activation functions, and initialization interact with one another?

Despite recent progress, these questions remain only partially understood. In this project, we focus specifically on the role of **depth** and **width** in neural networks.

Motivation. In Exercise 8 of the midterm, we studied a simple but important example: the triangle function

$$f_1(x) = \begin{cases} 2x, & 0 \le x \le \frac{1}{2}, \\ 2(1-x), & \frac{1}{2} \le x \le 1, \\ 0, & \text{otherwise}, \end{cases}$$

and its repeated compositions

$$f_{L+1}(x) = f_1(f_L(x)).$$

The key observation is that f_L has exactly 2^L linear regions on $[0, 1]$. By contrast, a one-dimensional ReLU network with a single hidden layer and width k has at most $k + 1$ linear regions. Therefore, representing f_L using a shallow network requires the width to grow exponentially with L.

Project Idea. The goal of this project is to investigate this phenomenon **from a numerical and experimental perspective**. In particular, the project asks:

> Can we observe experimentally that deep ReLU networks approximate highly oscillatory triangle functions more efficiently than shallow ReLU networks?

Suggested References. (The following papers are optional and intended only for further reading. If you already have a project idea, you do not need to read them.)

- M. Telgarsky, *Representation Benefits of Deep Feedforward Networks*, arXiv 2015.
- M. Telgarsky, *Benefits of Depth in Neural Networks*, COLT 2016.
- G. Montúfar, R. Pascanu, K. Cho, and Y. Bengio, *On the Number of Linear Regions of Deep Neural Networks*, NeurIPS 2014.
- R. Pascanu, G. Montúfar, and Y. Bengio, *On the Number of Response Regions of Deep Feed Forward Networks with Piecewise Linear Activations*, ICLR 2014.

7.2 Understanding the Muon Optimizer

The Muon optimizer is a recently developed optimization method for deep neural networks and large language models (LLMs), known for its fast convergence, improved training stability, and strong scalability in large-scale LLM training. It has been publicly discussed and adopted by organizations such as Moonshot AI and DeepSeek.

Possible project directions include:

- **Explore (or explain) one or more core mathematical ideas behind Muon or its variants** using the modded-nanogpt optimization repository. In particular, you may critically revisit the Muon Optimizer Blog, focusing on questions such as:
  - How does Muon use Newton–Schulz iterations? (This can be viewed as a continuation or extension of the bonus exercise from the midterm exam.)
  - Why are Newton–Schulz iterations useful for optimization?
- **Apply Muon or its variants to a learning task of your choice**, and compare them experimentally with SGD or AdamW in terms of some of the following metrics:
  - training loss;
  - convergence speed;
  - training stability;
  - computational cost.

  You are also encouraged to experiment with different coefficients in the Newton–Schulz iterations and analyze their impact on optimization performance. For example, one interesting question is:

  > Why does DeepSeek use the hybrid Newton–Schulz coefficients
  >
  > $$(a, b, c) = (3.4445, -4.7750, 2.0315)?$$

Suggested References.

- Keller Jordan, *Muon: An Optimizer for Hidden Layers in Neural Networks*, 2024. Muon Optimizer Blog
- DeepSeek-V4 Technical Report, 2026. DeepSeek-V4 Report
- G. Kim and M. Oh, *Convergence of Muon with Newton–Schulz*, ICLR 2026. Convergence of Muon with Newton–Schulz
- Modded-NanoGPT Optimization Benchmark by Keller Jordan: Repository Link

7.3 Crisp Specifications as a New Core Skill

As emphasized by Jeff Dean (Chief AI Scientist at Google) in *Owning the AI Pareto Frontier* (2026), one increasingly important skill in the AI era is the ability to produce **crisp specifications**. In particular, Jeff Dean noted that, in the age of coding agents and AI-assisted development, people will increasingly need to be "good at crisply specifying things rather than leaving things underspecified or ambiguous."

The topic of your project is intentionally **open-ended** and driven by your own interests. You are encouraged to explore problems that you genuinely care about and develop your own ideas and directions.

At the same time, your project should include:

- a clear explanation of the motivation;
- a precise problem statement;
- a **consistent mathematical formulation**;
- meaningful theoretical and/or experimental results, supported by concrete and consistent methods and/or algorithms.

Your project does not need to completely solve a grand or real-world problem. Initial results, exploratory findings, negative results, or partial progress are all valuable. You should think of this project as a potential starting point for future research, experimentation, or development.

More importantly, the project is an opportunity to practice an increasingly important skill in modern AI workflows:

> clearly specifying goals, assumptions, constraints, methods, and evaluation criteria, instead of leaving key aspects ambiguous or underspecified.

7.4 Suggested Papers for Critical Review or Literature Survey (Papers Outside This List Are Also Welcome)

If you prefer to critically revisit a published paper, simply choose a paper that interests you and analyze its motivations, methods, assumptions, strengths, limitations, and possible future directions.

If you prefer to conduct a literature survey, you may instead choose a machine learning topic of interest and summarize recent developments in that area.

- *Deep Learning*, Yann LeCun, Yoshua Bengio, and Geoffrey Hinton, Nature (2015).
- *Discovering Faster Matrix Multiplication Algorithms with Reinforcement Learning*, Alhussein Fawzi et al., Nature (2022).
- *An Overview of Gradient Descent Optimization Algorithms*, Sebastian Ruder, arXiv (2017).
- *A ConvNet for the 2020s*, Zhuang Liu et al., CVPR (2022).
- *What Is Local Optimality in Nonconvex–Nonconcave Minimax Optimization?*, Chi Jin, Praneeth Netrapalli, and Michael Jordan, ICML (2020).
- *Moreau Envelope for Nonconvex Bi-Level Optimization: A Single-Loop and Hessian-Free Solution Strategy*, Liu et al., ICML (2024).
- *Auto-Encoding Variational Bayes*, Diederik Kingma and Max Welling, ICLR (2014).

8 Getting Started

Finding Your Project Idea

1) Start with the Project Ideas
2) Use AI tools to explore and refine your ideas
3) Take time to find a topic that you genuinely care about

AI tools are becoming an essential part of modern scientific and engineering workflows. **Learning how to work effectively with AI systems is an important skill.**

> *"Practice working with AI. Human + AI will be superior to human or AI alone for the foreseeable future. Those who can work most effectively with AI will be the most highly valued."*
>
> — *Noam Brown (OpenAI)*

9 Project Recommendations

- Start with a minimal working version before attempting more ambitious ideas.
- Test and validate your methods and results carefully.
- Document your development process, experiments, and observations clearly.
- Use AI tools to explore alternative ideas, implementations, and research directions.
- Test your implementation incrementally as you develop it.
- Run small-scale experiments before scaling up to larger settings.

10 Writing and Presentation Language

- The final report and slides should be written in English. You are encouraged to use AI tools to improve the quality of your writing. For example, you may use prompts such as:

  > "Below is a paragraph from an academic paper. Polish the writing to meet academic style standards, improving spelling, grammar, clarity, concision, and overall readability. When necessary, rewrite the entire sentence. First provide the polished paragraph, then list all modifications and explain the reasons for each change in a markdown table."

- The presentation video may be given in English, Chinese, or a combination of both.